

7) Project summary

Triple-negative breast cancers (TNBC) face a lack of therapeutic options and a high mortality rate. Metaplastic breast carcinomas (MpBC) are a rare subtype of TNBC, characterized by high cellular plasticity and the presence of at least one transdifferentiated tumor compartment. These compartments can be of different non-epithelial types, the presence of which is exclusively determined histologically, and the mechanisms underlying their occurrence remain unexplained. Due to their rarity and complexity, the biology of MpBCs is indeed poorly understood and patients are still poorly managed.

The objectives of the project are to better understand the emergence of transdifferentiated compartments in MpBCs, and to provide molecular diagnostic options for these tumors. The analyses will focus on MpBCs of mixed histology, in which it is possible to jointly study pairs of phenotypically different compartments using multi-omic technologies: spatial & single-cell transcriptomics; exome sequencing and methylome after microdissection. This will allow a better understanding of the origin of different types of transdifferentiation (spindle, squamous, chondroid, osteosarcomatoid) and the genetic / non-genetic mechanisms underlying them. The availability of single-cell data will also make it possible to precisely determine the genes and signaling pathways that are specifically deregulated in transdifferentiated cancer cells, both compared to classic epithelial cancer cells and to the normal mesenchymal cells that they histologically resemble. This will also make it possible to identify molecular diagnostic solutions for these cancers, which are still poorly treated.

To achieve its objective, the project is structured around three specific axes:

- 1) Creation of a transcriptomic atlas dedicated to MpBCs with single-cell and spatial resolution
- 2) Characterization of the genetic and non-genetic determinants of transdifferentiation in MpBCs.
- 3) Identification of expression markers and signaling pathways specific to each subtype.